

Lubomír Bureš

Nuclear engineer focused on computational modelling and technical problem solving.

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Professional Experience

Saltfoss Energy (formerly Seaborg Technologies)

Copenhagen, Denmark

Senior Multiphysics Specialist

March 2024 – present

- Collaborate as a technical lead on the Saltfoss molten-salt reactor design, guiding key architectural decisions and ensuring effective system integration.
- Address complex technical challenges in reactor physics, thermal hydraulics, and design optimisation.
- Develop, verify, and validate computational models for advanced nuclear applications.
- Publish research and present at scientific conferences to bolster the company's technical visibility.

Multiphysics Specialist

October 2021 – March 2024

- Developed multiple computational models for molten-salt reactor analysis.
- Supervised a master thesis project with the candidate then joining the company as a full-time employee.
- Published peer-reviewed research and represented the company both in Denmark and internationally.

Paul Scherrer Institute

Villigen, Switzerland

Doctoral Researcher

October 2018 – September 2021

- Conducted independent doctoral research in computational physics.
- Designed and programmed software for high-fidelity numerical simulations applied to scientific research.
- Supervised four student projects, resulting in published research and successful student outcomes.
- Led exercise sessions for a course at ETH Zurich, improving student understanding.

Supplementary Projects

Winterlead

Remote

Independent Consultant

April 2024 – present

- Provide software development, computational engineering, and consultancy services on an ad hoc basis.
- Developed software solutions for Quantified Carbon.
- Volunteered technical expertise to the Repower Initiative.

Education

EPFL, Institute of Physics

Paul Scherrer Institute (PSI), Villigen, Switzerland

Doctor of Science (PhD)

2018 – 2021

- *Degree awarded by:* École Polytechnique Fédérale de Lausanne (EPFL)
- *Thesis:* Fundamental study on microlayer dynamics in nucleate boiling
- *Achievements:* Passed without reserve (highest grade). Awarded the 2022 PhD Prize by the European Nuclear Education Network Association. Awarded the 2019 Fellowship for Research in Japan (cancelled due to Covid-19) by the Japanese Society for the Promotion of Science.

ETH Zurich – EPF Lausanne
Joint MSc in Nuclear Engineering

Zürich & Lausanne, Switzerland
2016 – 2018

- *Thesis*: Direct numerical simulation of phase change in the presence of non-condensable gases
- *Achievements*: Graduated with distinction, overall GPA: 3.97/4.00 (third best among 1043 Master graduates). Awarded the EPFL Excellence Fellowship over the duration of the study programme.
- *Certifications*: Received the European Master of Science in Nuclear Engineering certification.
- *Internships*: National Cooperative for the Disposal of Radioactive Waste (Wettingen, Switzerland).
- *Extracurricular*: Represented the class in the EPFL student organisation (Fall semester 2016).

Czech Technical University
BSc in Nuclear Engineering

Prague, Czech Republic
2013 – 2016

- *Thesis*: Application of calculation codes based on Monte-Carlo method for benchmark experiments
- *Achievements*: Graduated with honours, overall GPA: 3.94/4.00. Won a student thesis competition. Received the 2015 Talented Student Award from the Czech Technical University.
- *Internships*: Nuclear Physics Institute of the Czech Academy of Sciences (Řež, Czech Republic), Joint Institute for Nuclear Research (Dubna, Russia).

Skills

Languages	Czech native, English C2, German C1, Danish B2: Prøve i Dansk 3 – 8.5/12.
Programming	Python, C++, Modelica, MATLAB
Software Tools	Git, Jupyter, VS Code, Vim, MS Office, LaTeX
Operating Systems	Windows, Linux

Organisational Involvement

IDA Nuclear
Board Member

Danish Society of Engineers, Denmark
September 2025 – present

- Organise events promoting knowledge and education about nuclear and atomic technology.

Kildevæld Kulturcenter, Digitalt Værksted
Volunteer / Superuser

Copenhagen, Denmark
October 2025 – present

- Support open workshop sessions by helping citizens use digital fabrication tools, including 3D printing and laser cutting, and by sparring on project ideas.

PhD and Postdoc Association
President

Paul Scherrer Institute, Switzerland
October 2019 – May 2021

- Oversaw the creation of the association's bylaws, enabling its registration as a non-profit organisation.
- Led the association through the challenging times of the Covid-19 pandemic, ensuring continued support and engagement for members.
- Represented the association externally and within the institute, enhancing its visibility and influence.
- March 2019 to September 2019: Social Committee Head.

Various
Volunteer / Event Organiser

Europe-wide
October 2018 – November 2023

- Volunteered for and helped organise multiple climate-change-related events across Europe.
- Represented the Nuclear for Climate initiative at the UN Climate Change Conference COP25 in Madrid.